Business Findings Report

1. Executive summary

This project analyses sample Google Merchandise Store activity using Google Analytics 4 (GA4) event data in BigQuery.

The objective is to understand website activity, shopping behaviour, product performance, purchase revenue, device contribution, monthly revenue patterns and first-acquisition traffic sources.

The results show total recorded purchase revenue of **$362,165**, with desktop contributing the largest revenue share, Apparel being the highest-revenue product category, and Google being the largest reported first-acquisition source by purchase revenue.

2. Business questions

1. Which website activities happen most frequently?
2. How many users perform key shopping events such as product views, add-to-cart, checkout and purchase?
3. Which device category contributes the most purchase revenue?
4. Which individual products and product categories generate the highest revenue?
5. Which month generated the highest purchase revenue in the selected period?
6. Which first-acquisition traffic sources are associated with the most purchasing users and purchase revenue?

3. Dataset and scope

- **Source table:** `bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*`
- **Source platform:** Google BigQuery public dataset
- **Data type:** GA4 event-level e-commerce data
- **Period filtered in the queries:** 1 November 2020 to 31 January 2021
- **Analysis focus:** event activity, shopping events, purchase revenue, devices, products, categories, months and traffic sources.

4. Key findings

4.1 Website activity

The event summary identified the following high-volume activities:

Event	Total events	Unique users
page_view	1,350,428	269,792
user_engagement	1,058,721	213,004
scroll	493,072	138,098
view_item	386,068	61,252
session_start	354,970	267,116
add_to_cart	58,543	12,545
begin_checkout	38,757	9,715
purchase	5,692	4,419

Interpretation: Page views and engagement events are common, while purchase events are much less frequent. Event totals count activity occurrences; unique users count distinct pseudonymous user identifiers for each event.

4.2 Purchase revenue by device

Device	Purchase events	Purchasing users in device group	Revenue (USD)	Revenue share
Desktop	3,226	2,541	$208,815	57.66%
Mobile	2,355	1,851	$146,768	40.52%
Tablet	111	97	$6,582	1.82%
Total	**5,692**	—	**$362,165**	**100.00%**

Interpretation: Desktop generated the largest share of recorded purchase revenue. Mobile also contributed a substantial share.

4.3 Top products by revenue

Product	Units sold	Revenue (USD)
Google Zip Hoodie F/C	273	$13,788
Google Crewneck Sweatshirt N…	236	$10,714
Google Men's Tech Fleece Grey	134	$9,965
Google Badge Heavyweight Pull…	201	$9,712
Super G Unisex Joggers	308	$9,529

The query returned the top 10 products; the table above highlights five visible leading results.

Interpretation: The product with the highest revenue was not the product with the highest units sold. For example, the Google Zip Hoodie F/C generated $13,788 from 273 units, while Super G Unisex Joggers sold 308 units and generated $9,529. This demonstrates why both sales volume and revenue are useful product measures.

4.4 Revenue by product category

Category label in source data	Units sold	Revenue (USD)
Apparel	5,447	$171,727
New	2,132	$25,813
Rags	1,053	$23,860

Interpretation: Apparel was the highest-revenue category in the query results, contributing approximately 47.42% of total purchase revenue.

4.5 Monthly purchase revenue

Month	Purchase events	Purchasing users in month	Revenue (USD)
November 2020	2,054	1,532	$144,260
December 2020	2,434	1,975	$160,555
January 2021	1,204	1,069	$57,350
Total	**5,692**	—	**$362,165**

Interpretation:

- December generated the highest revenue: $160,555.
- Revenue increased by approximately 11.29% from November to December.
- January revenue was approximately 64.28% lower than December.

4.6 Revenue by first-acquisition traffic source

Traffic source label	Purchase events	Purchasing users in source group	Revenue (USD)
google	1,620	1,377	$104,831
<Other>	1,290	1,063	$81,099
(direct)	1,251	1,054	$79,650
(data deleted)	830	680	$50,064
shop.googlemerchandisestore…	701	568	$46,521
Total	**5,692**	—	**$362,165**

Interpretation:

- `google` was the largest reported source by purchase revenue, at approximately 28.95% of the total.
- The `(direct)` source contributed $79,650.